

Buyer Segmentation Project Using K-Means Clustering

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Technology Used:

- Python
 - Pandas
 - Scikit-Learn
 - Matplotlib
 - K-Means Clustering
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Objective

The goal of this project is to segment real estate buyers into different groups based on their characteristics and purchasing behavior.

Benefits

- Understand customer behavior
 - Identify high-value customers
 - Improve marketing strategies
 - Support business decision-making
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Datasets Used

Clients Dataset

- Client ID
- Age
- Gender
- Country
- Satisfaction Score
- Loan Applied

Properties Dataset

- Sale Price
- Floor Area
- Property Type
- Transaction Details

Total Records

- Clients: 2,000
 - Properties: 10,000
 - Merged Records: 7,305
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Steps Performed

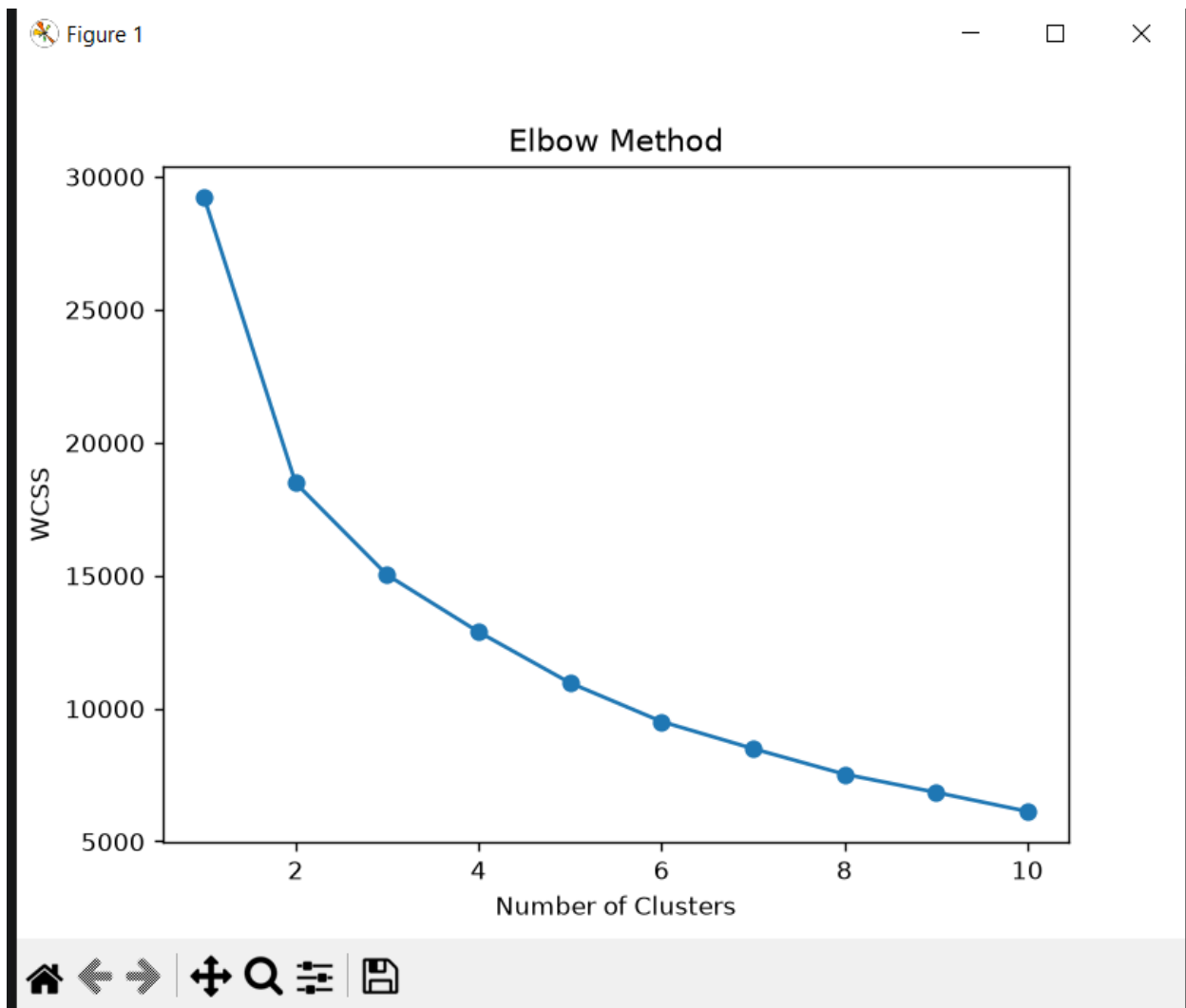
- ✓ Data Cleaning
- ✓ Missing Value Handling
- ✓ Age Calculation
- ✓ Sale Price Conversion
- ✓ Dataset Merging
- ✓ Feature Scaling

Features Used

- Age
 - Sale Price
 - Floor Area
 - Satisfaction Score
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Determining Optimal Number of Clusters

The Elbow Method was used to identify the best value of K.



Observation

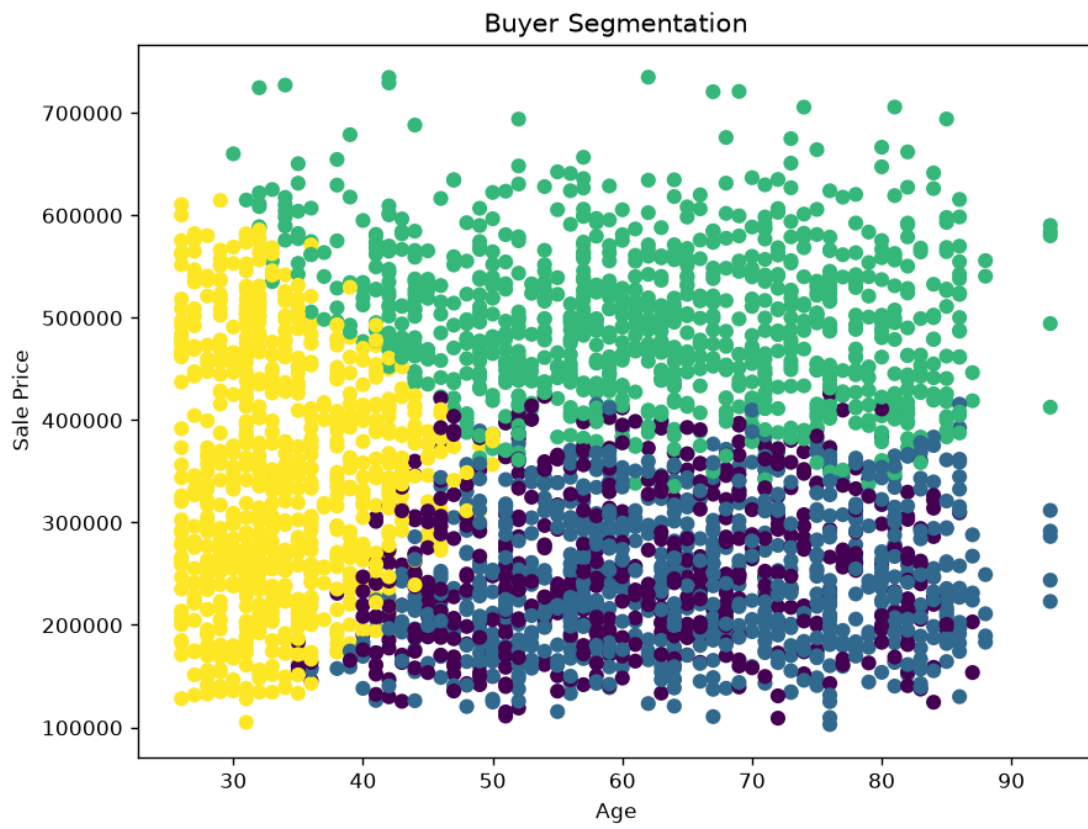
The curve bends around $K = 4$.

Therefore:

Optimal Number of Clusters = 4

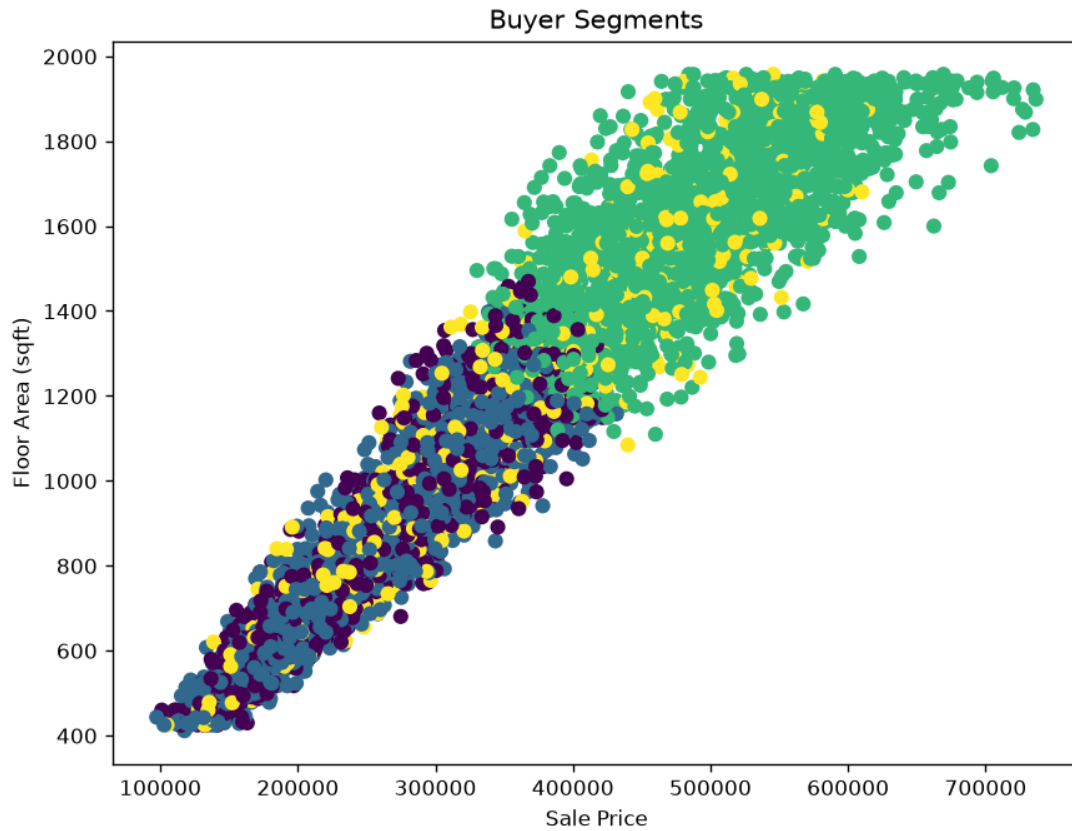
K-Means Clustering Output

Figure 1



(x, y) = (89.03, 5.29e+05)

Figure 1



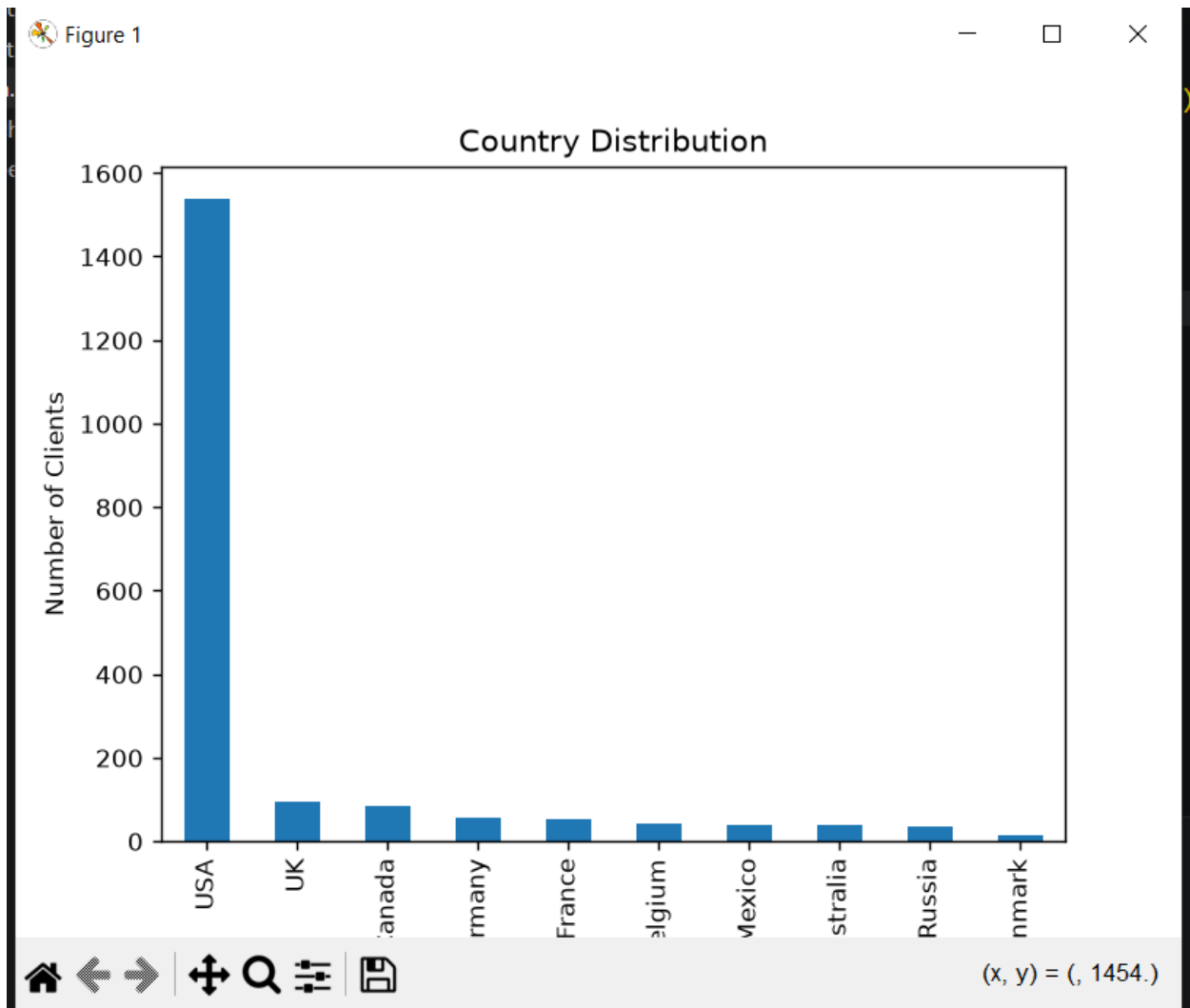
(x, y) = (6.851e+05, 1348.)

Key Findings

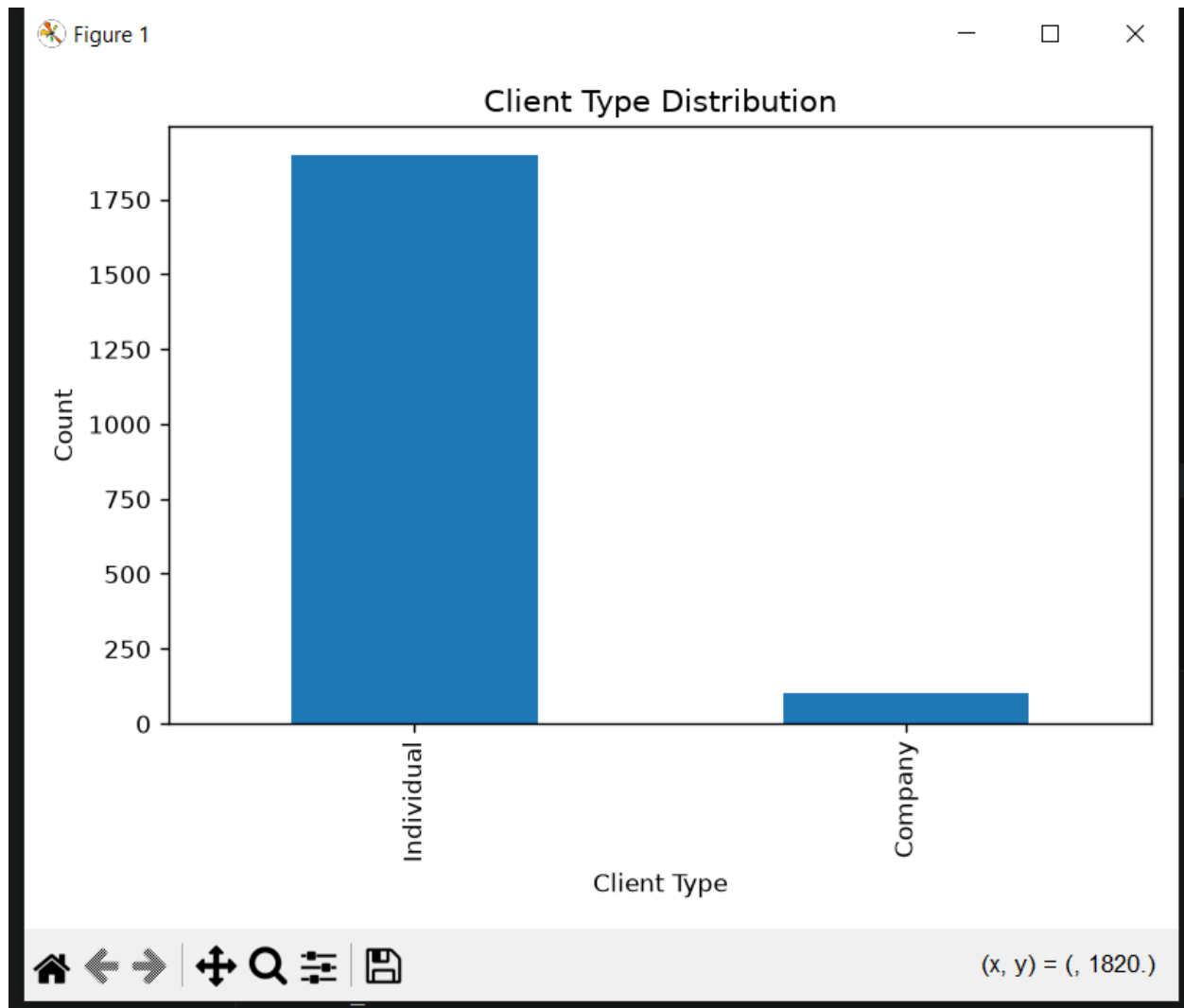
- Premium buyers spend more on larger properties.
- Younger buyers belong to a different segment.
- Four distinct customer groups were identified.

Client Type Distribution

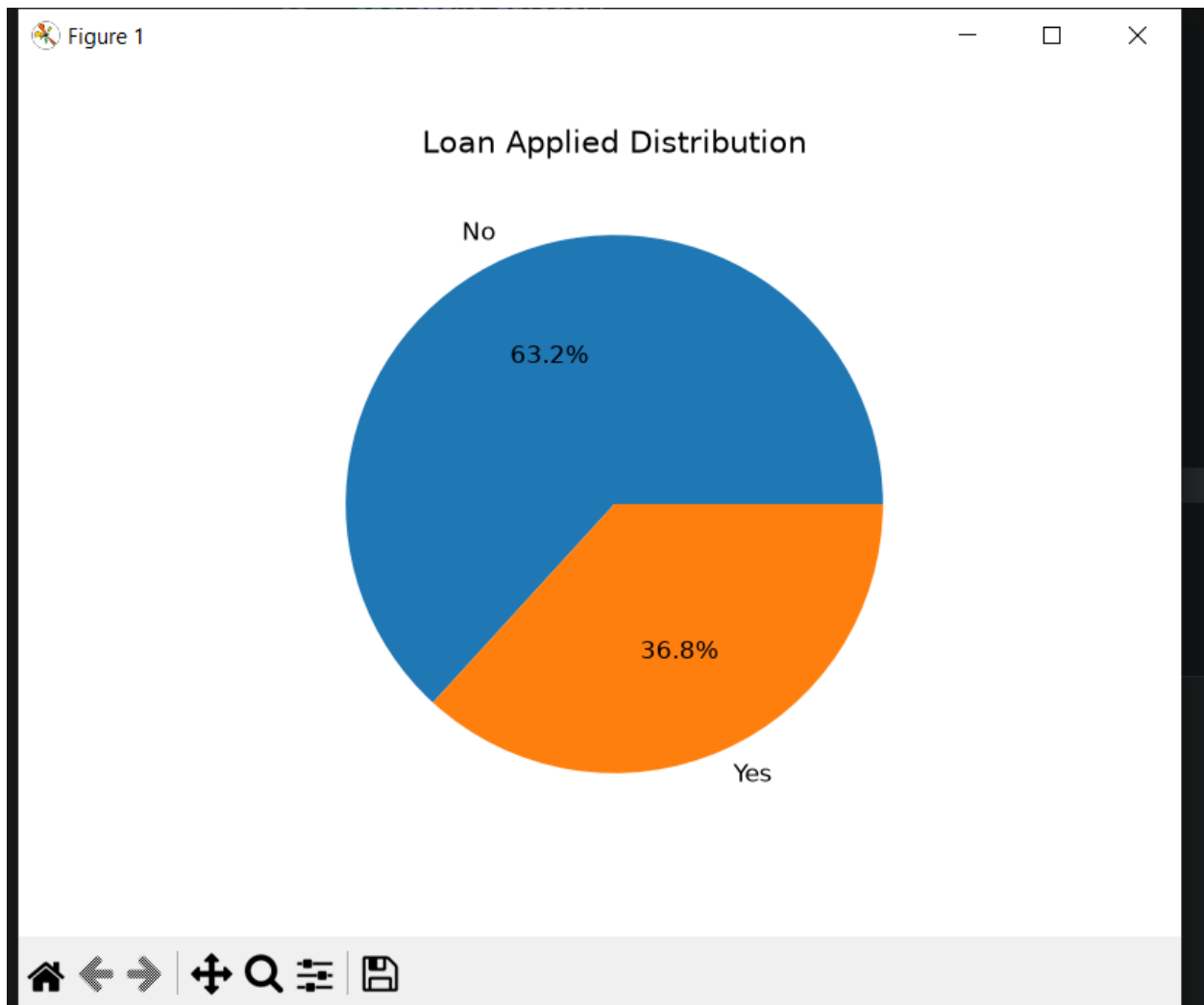
Country Distribution



Loan Applied Distribution



Pie Chart



Insights

- Most buyers are Individuals.
- USA has the highest number of customers.
- Majority of buyers did not apply for loans.

Cluster Summary

<u>Cluster</u>	<u>Description</u>
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Cluster 0	Older buyers, lower spending
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Cluster 1 Older buyers, high satisfaction

Cluster 2 Premium buyers, highest
spending

Cluster 3 Younger buyers, medium
spending

Final Conclusion

- Successfully segmented 7,305 buyers into 4 groups.
- K-Means clustering provided meaningful customer insights.
- Results can help improve marketing and sales strategies.

Thank You